PRODUCT USER MANUAL

For Sea Level TAC - DUACS products

SEALEVEL_GLO_PHY_L3_NRT _008_044
SEALEVEL_EUR_PHY_L3_NRT _008_059
SEALEVEL_GLO_PHY_L3_MY_008_062
SEALEVEL_EUR_PHY_L3_MY_008_061
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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2025/05/28		Change Format file: subdivision of unique PUM for L3/L4 products in 3 documents. This new doc concern L3 products	K Grassi	M-I Pujol

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GLOSSARY AND ABBREVIATIONS

MFC	Monitoring and Forecasting Centre
TAC	Thematic Assembly Center
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
MFWAM	Meteo France WAve Model
NRT	Near Real Time
MY	Multi Year
PUM	Product User Manual
QUID	Quality Information Document
CMEMS	Copernicus Marine Environment Monitoring Service
NetCDF	Network Common Data Form
CF	Climate Forecast (convention for NetCDF)
MDS	Marine Data Store
FORECAST	An assessment of the future state of the ocean variables made using numerical models.
HINDCAST	An assessment of the past state of the ocean variables made using numerical models with or without data assimilation. Simulations and analyses are hindcasts.
SIMULATION	An hindcast done without data assimilation.
NRT-ANALYSIS:	An hindcast done with data assimilation for a period very close to present. The quality and the number of available observations it's not optimal (less observations and/or less accurate than in a best-analysis).
BEST-ANALYSIS:	An hindcast done with data assimilation for a period close to present. The quality and the number of available observations it's the best available (more observations and/or more accurate than in a NRT-analysis).

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

The Sea Level TAC (Thematic Assembly Centre) is one of the eight TAC of the Copernicus Marine Service. The aim of this document is to describe the products delivered by the Sea Level TAC.
The data produced in the frame of this TAC are generated by the processing system including data from all altimeter Copernicus missions (Sentinel-6A, Sentinel-3A/B) and other collaborative or opportunity missions (e.g. :Jason-3, Saral[-DP]/AltiKa, Cryosat-2, OSTM/Jason-2, Jason-1, Topex/Poseidon, Envisat, GFO, ERS-1/2, Haiyang-2A/B).
The SL-TAC main products lines consist in **Level-3 (L3; along-track) and Level-4 (L4; gridded) altimeter products**. Both are produced in **Near Real Time (NRT) or Delayed Time/ Multi-Year (DT/MY)**. Through the SL-TAC, the Copernicus Marine Service also disseminates L3 products/datasets produced under the Eumetsat responsibility. They mainly concern the Copernicus altimeter mission Sentinel-6. Additionally, **auxiliary/statics product** can be delivered for additional information on the L3/L4 products quality. We describe here after the different L3 along-track product lines.
The data provided to users have a **global coverage and regional products** are also computed over European Seas, including part of the North-East Atlantic, Baltic, Mediterranean and Black Seas.

a. INear Real time products

The user can check the version of the product used with the global attributes given in netCDF files:

- product_version : gives the information of the main Copernicus Marine release for which the last change has been implemented (e.g. "vDec2021")
- software_version : gives the precise software version used for the production, and information of the reference altimeter standards and processing used (e.g. : "19.0.0_DUACS_DT2021_baseline")

TThe lists the system's noticeable events, outage or changes, as well as details of the impact on product quality events can be found in the associated QUID document (see section V)

b. Multi-Year products

The user can check the version of the product used with the global attributes given in netCDF files:

- product_version : gives the information of the main release for which the last change has been implemented (e.g. "vNov2024")
- software_version : gives the precise software version used for the production, and information of the reference altimeter standards and processing used (e.g. : "8.0_DUACS_DT2024_baseline")

The lists the system's noticeable events, outage or changes, as well as details of the impact on product quality events can be found in the associated QUID document (see section "V.4: V.3 REP/DT sub-system version changes").

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	SEALEVEL_GLO_PHY_L3_NRT _008_044 SEALEVEL_EUR_PHY_L3_NRT _008_059 SEALEVEL_GLO_PHY_L3_MY_008_062 SEALEVEL_EUR_PHY_L3_MY_008_061
Geographical coverage	GLO : Global Ocean EUR : European Area (30.0625°W-42.0625°E ; 19.9375°N-66.0625°N)
Variables	The different variables included in this product are listed in section 1.2.1
Product Type	NRT : Near Real Time MY : Multi Year
Available time series	NRT : usually the last 2 years are available online for historic product/dataset lines. This period can be reduced at the start of production of a new product/dataset line MY : 13/10/1992; -5 to 9M before present.
Temporal resolution	Daily
Update Frequency	
Target delivery time	NRT: up to 10 times a day MY : ~4 times a year
Delivery mechanism	Marine Information System
Horizontal resolution	~7km (1Hz) for "*PT1S" datasets ~1.3km (5Hz) for "PT0.2S" datasets
Number of vertical levels	1 (sea surface)
Format	Netcdf CF1.6

b) Production System Description

The SEALEVEL altimeter products are processed with the CNES/CLS DUACS (Data Unification and Altimeter Combination System) system.
The DUACS processing steps are described in the QUID document.

The altimeter measurements used as input of the DUACS system consist in a majority of Level 2 products from different missions, that are available under three forms, with different delay of availability:

- Fast delivery or Near Real Time products (OGDR or NRT). These products do not always benefit from precise orbit determination, nor from some external model-based corrections (Dynamic Atmospheric Correction (DAC), Global Ionospheric Maps (GIM)).
- The Intermediate or Slow Time Critical products (IGDR or STC) that are the latest high-quality altimeter data produced in near-real-time
- Delayed Time or Non-Time Critical product (GDR or NTC).

Details of the different L2 altimeter products sources and delay of availability are given in the QUID document.

c) Dataset details

Different datasets are defined for the L3 products. They relate to the different altimeter missions processed. The following tables list the different datasets available for each product line and give an overview of the different physical variables available. Detailed information on the different altimeters processed is available on the QUID document.

C. L3 NRT 1Hz datasets

SEALEVEL_GLO_PHY_L3_NRT _008_044	**SEALEVEL_EUR_PHY_L3_NRT _008_059**
Dataset: cmems_obs-sl_glo_phy-ssh_nrt_al-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_nrt_c2n-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_nrt_h2b-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_nrt_J3-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_nrt_j3n-l3-duacs_PT1S* cmems_obs-sl_glo_phy-ssh_nrt_j3g-l3-duacs_PT1S-i cmems_obs-sl_glo_phy-ssh_nrt_s3a-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_nrt_s3b-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_nrt_s6a-hr-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_nrt_swon-l3-duacs_PT1S	**Dataset:** cmems_obs-sl_eur _phy-ssh_nrt_al-l3-duacs_PT1S cmems_obs-sl_eur _phy-ssh_nrt_c2n-l3-duacs_PT1S cmems_obs-sl_eur _phy-ssh_nrt_h2b-l3-duacs_PT1S cmems_obs-sl_eur _phy-ssh_nrt_J3-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_nrt_j3n-l3-duacs_PT1S* cmems_obs-sl_eur_phy-ssh_nrt_j3g-l3-duacs_PT1S-i cmems_obs-sl_eur_phy-ssh_nrt_s3a-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_nrt_s3b-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_nrt_s6a-hr-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_nrt_swon-l3-duacs_PT1S
Physical variables name in the NetCDF file and Unit: Long_name & Standard_name	
sla_unfiltered [m] Sea level anomaly not-filtered not-subsampled with dac, ocean_tide and lwe correction applied sea_surface_height_above_sea_level	
sla_filtered [m] Sea level anomaly filtered not-subsampled with dac, ocean_tide and lwe correction applied sea_surface_height_above_sea_level	
mdt [m] Mean dynamic topography sea_surface_height_above_geoid	
dac [m] Dynamic Atmospheric Correction -	

ocean_tide [m] Ocean tide model -
internal_tide [m] Internal Tide signal: coherent mode M2/K1/O1/S2 -
lwe [m] Long wavelength error -
Coordinate variables name in the NetCDF file and Unit: Long_name & Standard_name
time [days since 1950-01-01 00:00:00] Time of measurement time
longitude [degrees_east] Longitude of measurement (theoretical position for repetitive orbits) longitude
latitude [degrees_north] Latitude of measurement (theoretical position for repetitive orbits) latitude
Other useful variables name in the NetCDF file and Unit: Long_name & Standard_name
Flag [0,1] Data origin -
cycle [1] Cycle the measurement belongs to -
track [1] Track in cycle the measurement belongs to -

d. L3 NRT 5Hz datasets

SEALEVEL_GLO_PHY_L3_NRT _008_044	**SEALEVEL_EUR_PHY_L3_NRT _008_059**
Dataset: cmems_obs-sl_glo_phy-ssh_nrt_h2b-l3-duacs_PT0.2S cmems_obs-sl_glo_phy-ssh_nrt_j3n-l3-duacs_PT0.2S* cmems_obs-sl_glo_phy-ssh_nrt_j3g-l3-duacs_PT0.2S-i cmems_obs-sl_glo_phy-ssh_nrt_s3a-l3-duacs_PT0.2S cmems_obs-sl_glo_phy-ssh_nrt_s3b-l3-duacs_PT0.2S cmems_obs-sl_glo_phy-ssh_nrt_s6a-hr-l3-duacs_PT0.2S cmems_obs-sl_glo_phy-ssh_nrt_swon-l3-duacs_PT0.2S	**Dataset:** cmems_obs-sl_eur_phy-ssh_nrt_h2b-l3-duacs_PT0.2S cmems_obs-sl_eur_phy-ssh_nrt_j3n-l3-duacs_PT0.2S* cmems_obs-sl_eur_phy-ssh_nrt_j3g-l3-duacs_PT0.2S-i cmems_obs-sl_eur_phy-ssh_nrt_s3a-l3-duacs_PT0.2S cmems_obs-sl_eur_phy-ssh_nrt_s3b-l3-duacs_PT0.2S cmems_obs-sl_eur_phy-ssh_nrt_s6a-hr-l3-duacs_PT0.2S cmems_obs-sl_eur_phy-ssh_nrt_swon-l3-duacs_PT0.2S
Physical variables name in the NetCDF file and Unit: Long_name & Standard_name	
sla_unfiltered [m] Sea level anomaly not filtered and 5Hz subsampled with dac, ocean_tide, iw and lwe correction applied sea_surface_height_above_sea_level	
sla_filtered [m] Sea level anomaly low-pass filtered and 5Hz subsampled with dac, ocean_tide, iw and lwe correction applied sea_surface_height_above_sea_level	
mdt [m] Mean dynamic topography sea_surface_height_above_geoid	

dac [m] Dynamic Atmospheric Correction -
ib_lf [m] Low frequency part of inverse barometer sea_surface_height_correction_due_to_air_pressure_at_low_frequency
ocean_tide [m] Ocean tide model sea_surface_height_amplitude_due_to_geocentric_ocean_tide
internal_tide [m] Internal Tide signal: coherent mode M2/K1/O1/S2 -
lwe [m] Long wavelength error -
xtgosa [m.s-1] Anomaly of the geostrophic velocity in the across-track direction -
xtgosm [m.s-1] Mean geostrophic velocity in the across-track direction -
xtdir [degrees] Across-Track direction sea_water_velocity_to_direction
Coordinate variables name in the NetCDF file and Unit: Long_name & Standard_name
time [days since 1950-01-01 00:00:00] Time of measurement time
longitude [degrees_east] Longitude of measurement longitude
latitude [degrees_north] Latitude of measurement latitude
longitude_theoretical [degrees_east] Theoretical longitude of measurement (defined for repetitive orbits) longitude
latitude_theoretical [degrees_north] Theoretical latitude of measurement (defined for repetitive orbits) latitude
distance_from_theoretical [m] Distance between the real position and the theoretical position of measurement -
Other useful variables name in the NetCDF file and Unit: Long_name & Standard_name
flag [0,1] Data origin -
cycle [1] Cycle the measurement belongs to -

track [1] Track in cycle the measurement belongs to -

II.2.1.3 L3 MY 1Hz datasets

SEALEVEL_GLO_PHY_L3_MY_008_062	**SEALEVEL_EUR_PHY_L3_MY_008_061**
Dataset: cmems_obs-sl_glo_phy-ssh_my_alg-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_al-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_c2-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_c2n-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_e1g-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_e1-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_e2-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_enn-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_en-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_g2-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_h2ag-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_h2a-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_h2b-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_j1g-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_j1n-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_j1-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_j2g-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_j2n-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_j2-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_j3-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_j3n-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_j3g-l3-duacs_PT1S-i cmems_obs-sl_glo_phy-ssh_my_s3a-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_s3b-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_s6a-lr-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_tpn-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_tp-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_swonc-l3-duacs_PT1S cmems_obs-sl_glo_phy-ssh_my_swon-l3-duacs_PT1S	**Dataset:** cmems_obs-sl_eur_phy-ssh_my_alg-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_al-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_c2-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_c2n-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_e1g-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_e1-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_e2-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_enn-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_en-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_g2-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_h2ag-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_h2a-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_h2b-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_j1g-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_j1n-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_j1-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_j2g-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_j2n-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_j2-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_j3-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_j3n-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_j3g-l3-duacs_PT1S-i cmems_obs-sl_eur_phy-ssh_my_s3a-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_s3b-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_s6a-lr-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_tpn-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_tp-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_swonc-l3-duacs_PT1S cmems_obs-sl_eur_phy-ssh_my_swon-l3-duacs_PT1S
Physical variables name in the NetCDF file and Unit: Long_name & Standard_name	
sla_unfiltered [m] Sea level anomaly not-filtered not-subsampled with dac, ocean_tide and lwe correction applied sea_surface_height_above_sea_level	
sla_filtered [m] Sea level anomaly filtered not-subsampled with dac, ocean_tide and lwe correction applied sea_surface_height_above_sea_level	
mdt [m] Mean dynamic topography sea_surface_height_above_geoid	
dac [m] Dynamic Atmospheric Correction -	
ocean_tide [m] Ocean tide model -	
internal_tide [m] Internal Tide signal: coherent mode M2/K1/O1/S2 -	

lwe [m] Long wavelength error -
Coordinate variables name in the NetCDF file and Unit: Long_name & Standard_name
time [days since 1950-01-01 00:00:00] Time of measurement time
longitude [degrees_east] Longitude of measurement (theoretical position for repetitive orbits) longitude
latitude [degrees_north] Latitude of measurement (theoretical position for repetitive orbits) latitude
Other useful variables name in the NetCDF file and Unit: Long_name & Standard_name
tpa_correction [m] TOPEX-A instrumental drift correction derived from altimetry and tide gauges global comparisons (WCRP Sea Level Budget Group, 2018) sea_surface_height_above_sea_level
cycle [1] Cycle the measurement belongs to -
track [1] Track in cycle the measurement belongs to -
track [1] Track in cycle the measurement belongs to -

d) Additional information on some parameters

Sla **sla_filtered** **sla_unfiltered**	Sea Level Anomaly (see Annex) Unfiltered : raw measurement including noises Filtered: low-pass filtered SLA for noise reduction
xtgosa & xtgosm [m.s-1] **Xtdir [degrees]**	Anomalies of the geostrophic current and mean geostrophic current in the across-track direction. Across-track direction
Mdt [m]	Mean Dynamic Topography (see Annex)
Dac [cm]	This correction allows for the removal of high frequency variability induced by the atmospheric forcing and aliased by the altimetric measurements. The high frequency part is based on a barotropic model simulation forced by atmospheric pressure and winds (MOG2D; Carrère and Lyard 2003); the low frequency part is an inverse barometer response. A 20-day cutoff-period was chosen because it corresponds to the Nyquist period of T/P-Jason reference altimeters sampling and because the variability is mostly barotropic in this high frequency band. This correction is already included in the SLA so if the user wants to uncorrect it or to use another correction instead, he must add it to the SLA from the product:

	sla_filtereduncorrected = sla_filteredfrom product + dac. sla_unfiltereduncorrected = sla_unfilteredfrom product + dac.
Ib_lf [cm]	Low frequency (periods > 20 days) part of inverse barometer. This is the low frequency component of the DAC correction, induced by atmospheric pressure only.
Lwe [cm]	This correction allows correcting along track altimeter signals from long wavelengths errors remaining in the signal. LWE are defined to be orbit errors (very long spatial scales) and residual high-frequency signals (short time scale and large spatial scales); LWE are assumed to be uncorrelated between tracks and cycles. The LWE estimation is similar to the optimal interpolation technique described in Le Traon et al.[1998]; the main difference is that along-track LWE are estimated instead of the ocean signal. This correction is already included in the SLA but it is stored with opposite sign compared to the other corrections so if the user wants to uncorrect it or to use another correction instead, he must subtract it from the SLA in the product: sla_filtereduncorrected = sla_filteredfrom product - lwe. sla_unfiltereduncorrected = sla_unfilteredfrom product - lwe.
ocean_tide [cm]	The oceanic tide combines the ocean tide model and the loading tide model. The models are described in the QUID document This correction is already included in the SLA so if the user wants to uncorrect it or to use another correction instead, he must add it to the SLA from the product: sla_filtereduncorrected = sla_filteredfrom product + oceanic_tide. sla_unfiltereduncorrected = sla_unfilteredfrom product + oceanic_tide.
internal_tide [cm]	The internal tide contains to the coherent part of the baroclinic tide. The model used is described in the QID document. The sla in this file is already corrected for the internal_tide; the uncorrected sla can be computed as follows: sla_filtereduncorrected = sla_filteredfrom product + internal_tide. sla_unfiltereduncorrected = sla_unfilteredfrom product + internal_tide.
tpa_correction [m]	"NOT IMPLEMENTED YET: This correction is not yet available for the current product version. This field will be updated once this correction is producted. This variable can be added to the gridded SLA to correct for the observed instrumental drift during the lifetime of the TOPEX-A mission (the correction is null after this period). This is a global correction to be added a posteriori (and not before) on the global mean sea level estimate derived from the gridded sea level map. It can be applied at regional or local scale as a best estimate (better than no correction, since the regional variation of the instrumental drift is unknown). When available this variable can be added to the SLA to correct for the observed instrumental drift during the lifetime of the TOPEX-A mission (the correction is null after this period). This is a global mean correction to be added a posteriori (and not before) on the global mean sea level estimate derived from the sea level variable. It can be applied at regional or local scale as a best estimate (better than no correction, since the regional variation of the instrumental drift is unknown) Global_MSLTPA_drift_corrected(d) = Global_MSLdeduced from SLA from product(d) + tpa_correction(d). Where d is the date of the day considered

e) Processing information

The NRT L3 products are disseminated several times each day (see next section).
The MY L3 products are extended in time nearly 3 or 4 times per years, depending on the upstream data availability.

f) Catalogue updating

e. Update Time for Near real time production : Level-3 along-track products

As described in Figure 1 below, several runs of the SL-TAC chain occur each day.
The files, tagged with the production date of the day, are actually generated by a run that starts at 23:00 the previous day. This is considered the nominal run, combining IGDR or L2P STC and OGDR or L2 NRT data. Every day, this run produces along-track products within 8 to 12 hours after the last measurement.
Moreover, multiple secondary runs take place throughout the day, incorporating the latest available measurements (i.e., fast-delivery upstream). This allows the production of along-track files within approximately 3 to 5 hours after the last measurement. This approach was implemented to enable downloading of the latest available measurement at any time during the day.

The delivery data flow is illustrated in the figure below, with an example based on a real situation.
The consolidated data, shown in green, are derived from near-real-time input data (IGDR/STC) and will not be updated in future processing. They correspond to data produced with a 21-day delay. For instance, the nominal production tagged on 2024/04/22 (i.e., the first line in the figure) corresponds to the delivery of data covering the period from 2024/04/01 at 00:00 to around midday on 2024/04/21 (depending on the satellite considered). In this case, measurements from 2024/04/01 are considered consolidated.
The data shown in yellow are computed using near-real-time input data and correspond to the time period that will be redelivered in subsequent runs until they become part of the consolidated period (i.e., green data). In the example provided in the figure, measurements from 2024/04/02 and 2024/04/03 are redelivered in the run tagged on 2024/04/23 (the last line in the figure). Additionally, measurements from 2024/04/03 are redelivered in the nominal runs tagged from 2024/04/24 onward (not visible in the figure).
The data shown in blue are generated using fast-delivery input data (OGDR/NRT). They generally cover one or two days before the run start date, depending on the mission considered. New measurements available throughout the day are ingested in various runs launched at different times, increasing the number of measurements accessible to users. The nominal run of the following day replaces these measurements with near-real-time input data (IGDR/STC) when available. In the example in the figure, measurements from 2024/04/21 are progressively completed during various runs on 2024/04/22 (the first four lines in the figure). They are then replaced by IGDR/STC measurements in the nominal run of 2024/04/23 (the last line in the figure).
Both global and regional products are available for a maximum historical period of two years. This period may be temporarily reduced to a minimum of 20 days following the implementation of product updates affecting the homogeneity of the time series (e.g., format changes, added/removed variables).

Both Global and Regional products are available over a maximal temporal period corresponding to the past 2 years. This period can be reduced to a minimal 20 days period after the implementation of a product evolution affecting the homogeneity of the time series (e.g. product format change; add/rm variables).

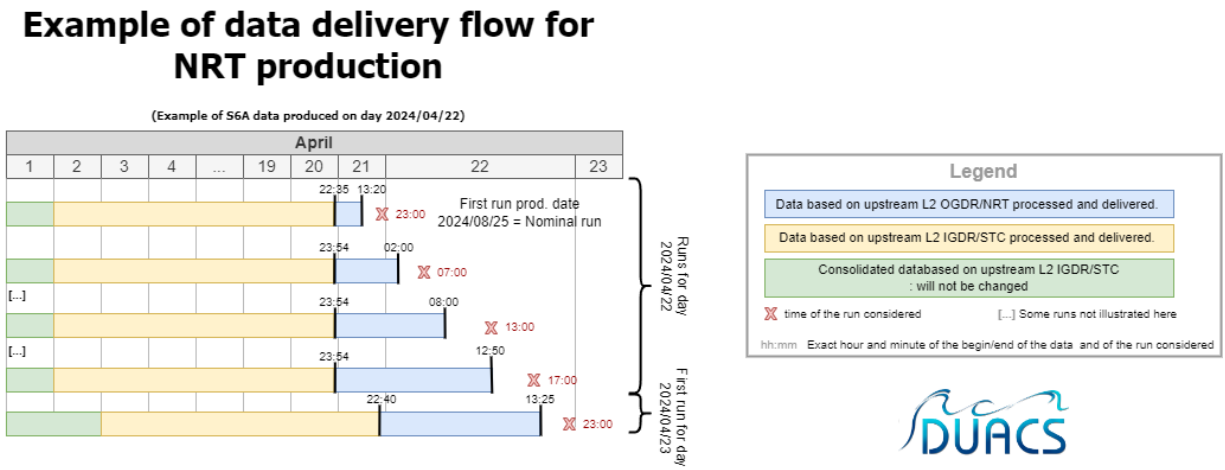

Figure 1: Example of data delivery flow for Global and regional NRT SL-TAC products

f. Update Time for Multi-Year production

The products/datasets are extended in time nearly four times a year in the nominal case, so that these datasets cover from 1993 up to five to nine months before present. These datasets benefit from the highest quality altimeter measurements and geophysical corrections, and are produced with a unique system (no changes in time in the set up, extended in time using the same system) to minimize the risk of quality loss or spurious signals appearing in time.
A full reprocessing of the Multi-Year series is usually delivery each 3 to 4 years.

g) Time averaging

L3 product are instantaneous measurement. There is no temporal mean version of the product.

h) Product use and impact

The L3 DUACS products can be used for different applications. We give here some examples:

- Scientific studies regarding physical and biogeochemical domains (see for instance DUACS studies and applications).
- Reconstruction of global/regional maps of sea surface height and associated geostrophic currents
- Assimilation in numerical physical ocean models

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

Level-3 along-track products

The nomenclature used is:

<delay>_<zone>_<mission>_<variable>_l3_<resolution>_<date>_<dateprod>.<format>

where the fields in "<>" are described below:

delay	nrt dt	near-real time products delayed time products
zone	global europe	global geographic coverage product Europe products
mission	e1 e1g e2 tp tpn g2 j1 j1n j1g j2 j2n j2g j3 j3n j3g	ERS-1 (only for dt) ERS-1 geodetic phase (only for dt) ERS-2 (only for dt) TOPEX/Poseidon (only for dt) TOPEX/Poseidon on its new orbit (only for dt) GFO (only for dt) Jason-1 (only for dt) Jason-1 on its new orbit (only for dt) Jason-1 on its geodetic orbit (only for dt) OSTM/Jason -2 (only for dt) OSTM/Jason -2 on its interleaved orbit OSTM/Jason-2 on its long repeat orbit (LRO) Jason-3 Jason-3 on its new (interleaved) orbit Jason-3 on its new (interleaved long repeat orbit) orbit

	en enn c2 c2n al alg h2a h2ag h2b s3a s3b s6a-hr s6a-lr swon swonc	Envisat (only for dt) Envisat on its new orbit (only for dt) Cryosat-2 Cryosat-2 on its new orbit Saral/AltiKa Saral/AltiKa on its geodetic orbit (only for dt) HaiYang-2A (only for dt) HaiYang-2A on its geodetic orbit (only for dt) HaiYang-2B Sentinel-3A Sentinel-3B Sentinel-6A with SAR mode measurement Sentinel-6A with LRM mode measurement SWOT-nadir SWOT-nadir (CALVAL phase with 1 day cycle, only for dt)
variable	phy	Ex: sla, mdt and some corrections
resolution	1hz 5hz	Product with 1Hz (~7km) posting rate Product with 5Hz (~1km) posting rate
date	YYYYMMDD	date of the dataset For NRT products : the file contains the measurements dated exactly on the day considered. This means that the altimeter tracks, for which the first measurement is dated at the end of a day and the last one at the beginning of the next day, are stored on two successive files For MY products: the whole altimeter track is stored in a unique file. This means that the file containing measurements of one day can also contain part of the measurements for the end of the previous day or beginning of the next day.
dateprod	YYYYMMDD	production date of the dataset
format	.nc	NetCdf CF1.6

4.1 FILE SIZE

The following table give an estimation of the typical size of one the daily file for L3 and L4 products, when downloaded through the Copernicus Marine FTP Service.

Area	Product Level	NetCDF file size
Global	L3 1Hz	900 kB

	L3 5Hz (nrt only)	5 to 6 MB
Europe	L3 1Hz	90 to 120 kB
	L3 5Hz (nrt only)	450 kB

4.2 STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5) REFERENCES

Quality Information Document CMEMS-SL-QUID-008-032-068.pdf

6) ANNEX

a) Level-3 along-track products

An example output NetCDF file header for the dataset-duacs-nrt-global-s6a-hr-phy-l3 dataset is given bellow.

```
netcdf nrt_global_s6a_hr_phy_l3_1hz_20240430_20240521 {
dimensions:
        time = 52300 ;
variables:
        double time(time) ;
                time:_FillValue = NaN ;
                time:axis = "T" ;
                time:long_name = "Time of measurement" ;
                time:standard_name = "time" ;
                time:units = "days since 1950-01-01T00:00:00+00:00" ;
                time:calendar = "gregorian" ;
        int longitude(time) ;
                longitude:add_offset = 0. ;
                longitude:long_name = "Longitude of measurement" ;
                longitude:scale_factor = 1.e-06 ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
        int latitude(time) ;
                latitude:add_offset = 0. ;
                latitude:long_name = "Latitude of measurement" ;
                latitude:scale_factor = 1.e-06 ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
        short cycle(time) ;
                cycle:coordinates = "longitude latitude" ;
                cycle:long_name = "Cycle the measurement belongs to" ;
                cycle:units = "1" ;
        short track(time) ;
                track:coordinates = "longitude latitude" ;
                track:long_name = "Track in cycle the measurement belongs to" ;
                track:units = "1" ;
        short sla_filtered(time) ;
                sla_filtered:_FillValue = 32767s ;
                sla_filtered:add_offset = 0. ;
                sla_filtered:comment = "The sea level anomaly is the sea surface height above mean sea surface height; the
uncorrected sla can be computed as follows: [uncorrected sla]=[sla from product]+[dac]+[ocean_tide]+[internal_tide]-[lwe]; see the
product user manual for details" ;
                sla_filtered:coordinates = "longitude latitude" ;
                sla_filtered:long_name = "Sea level anomaly low-pass filtered and 1Hz subsampled with dac, ocean_tide,
internal_tide and lwe correction applied" ;
                sla_filtered:scale_factor = 0.001 ;
                sla_filtered:standard_name = "sea_surface_height_above_sea_level" ;
                sla_filtered:units = "m" ;
        short sla_unfiltered(time) ;
                sla_unfiltered:_FillValue = 32767s ;
                sla_unfiltered:add_offset = 0. ;
                sla_unfiltered:comment = "The sea level anomaly is the sea surface height above mean sea surface height;
the uncorrected sla can be computed as follows: [uncorrected sla]=[sla from product]+[dac]+[ocean_tide]-[lwe]; see the product
user manual for details" ;
                sla_unfiltered:coordinates = "longitude latitude" ;
                sla_unfiltered:long_name = "Sea level anomaly not-filtered and 1Hz subsampled with dac, ocean_tide,
internal_tide and lwe correction applied" ;
                sla_unfiltered:scale_factor = 0.001 ;
                sla_unfiltered:standard_name = "sea_surface_height_above_sea_level" ;
                sla_unfiltered:units = "m" ;
        short dac(time) ;
                dac:_FillValue = 32767s ;
                dac:add_offset = 0. ;
                dac:comment = "The sla in this file is already corrected for the dac; the uncorrected sla can be computed as
follows: [uncorrected sla]=[sla from product]+[dac]; see the product user manual for details" ;
                dac:coordinates = "longitude latitude" ;
                dac:long_name = "Dynamic Atmospheric Correction" ;
                dac:scale_factor = 0.001 ;
                dac:units = "m" ;
```

```
        short ocean_tide(time) ;
                ocean_tide:_FillValue = 32767s ;
                ocean_tide:add_offset = 0. ;
                ocean_tide:comment = "The sla in this file is already corrected for the ocean_tide; the uncorrected sla can be
computed as follows: [uncorrected sla]=[sla from product]+[ocean_tide]; see the product user manual for details" ;
                ocean_tide:coordinates = "longitude latitude" ;
                ocean_tide:long_name = "Ocean tide model" ;
                ocean_tide:standard_name = "sea_surface_height_amplitude_due_to_geocentric_ocean_tide" ;
                ocean_tide:scale_factor = 0.001 ;
                ocean_tide:units = "m" ;
        short internal_tide(time) ;
                internal_tide:_FillValue = 32767s ;
                internal_tide:add_offset = 0. ;
                internal_tide:comment = "The sla in this file is already corrected for the internal_tide; the uncorrected sla can
be computed as follows: [uncorrected sla]=[sla from product]+[internal_tide]; see the product user manual for details" ;
                internal_tide:coordinates = "longitude latitude" ;
                internal_tide:long_name = "Internal Tide signal: coherent mode M2/K1/O1/S2" ;
                internal_tide:scale_factor = 0.001 ;
                internal_tide:units = "m" ;
        short lwe(time) ;
                lwe:_FillValue = 32767s ;
                lwe:add_offset = 0. ;
                lwe:comment = "The sla in this file is already corrected for the lwe; the uncorrected sla can be computed as
follows: [uncorrected sla]=[sla from product]-[lwe]; see the product user manual for details" ;
                lwe:coordinates = "longitude latitude" ;
                lwe:long_name = "Long wavelength error" ;
                lwe:scale_factor = 0.001 ;
                lwe:units = "m" ;
        short mdt(time) ;
                mdt:_FillValue = 32767s ;
                mdt:add_offset = 0. ;
                mdt:comment = "The mean dynamic topography is the sea surface height above geoid; it is used to compute
the absolute dynamic tyopography adt=sla+mdt" ;
                mdt:coordinates = "longitude latitude" ;
                mdt:long_name = "Mean dynamic topography" ;
                mdt:scale_factor = 0.001 ;
                mdt:standard_name = "sea_surface_height_above_geoid" ;
                mdt:units = "m" ;
        short flag(time) ;
                flag:_FillValue = 32767s ;
                flag:comment = "The origin of the data is determined by the type of geophysical data records (GDR) used in
computation of the SLA: 1 for an Interim GDR (IGDR) or Short Time Critical (STC) data and 0 for an Operational GDR (OGDR) or
Near Real Time (NRT) data" ;
                flag:coordinates = "longitude latitude" ;
                flag:long_name = "Data origin" ;
                flag:meaning = "OGDR_or_NRT, IGDR_or_STC" ;
                flag:units = 1LL ;
                flag:values = "0s, 1s" ;

// global attributes:
                :Conventions = "CF-1.6" ;
                :Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
                :cdm_data_type = "Swath" ;
                :comment = "Sea Surface Height measured by altimeters referenced to the [1993, 2012] period" ;
                :contact = "servicedesk.cmems@mercator-ocean.eu" ;
                :creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
                :creator_name = "CMEMS - Sea Level Thematic Assembly Center" ;
                :creator_url = "http://marine.copernicus.eu" ;
                :date_created = "2024-05-21T00:23:48Z" ;
                :date_issued = "2024-05-21T00:23:48Z" ;
                :date_modified = "2024-05-21T00:23:48Z" ;
                :geospatial_lat_max = 66.144886 ;
                :geospatial_lat_min = -66.149562 ;
                :geospatial_lat_resolution = 0.000259999999997262 ;
                :geospatial_lat_units = "degrees_north" ;
                :geospatial_lon_max = 179.999832 ;
                :geospatial_lon_min = -179.995425 ;
                :geospatial_lon_resolution = 0.032346000000004 ;
                :geospatial_lon_units = "degrees_east" ;
                :geospatial_vertical_max = 0. ;
                :geospatial_vertical_min = 0. ;
                :geospatial_vertical_positive = "down" ;
                :geospatial_vertical_resolution = "point" ;
                :geospatial_vertical_units = "m" ;
```

:history = "2024-05-21T00:23:48Z: Creation" ;
:institution = "CLS, CNES" ;
:keywords = "Oceans > Ocean Topography > Sea Surface Height" ;
:keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
:license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
:platform = "Sentinel-6A" ;
:processing_level = "L3" ;
:product_version = "vNov2023" ;
:project = "EUMETSAT Sentinel L2P/L3 Marine Altimetry Service" ;
:references = "http://marine.copernicus.eu" ;
:software_version = "2.0.0_DUACS_DT2021_baseline" ;
:source = "Sentinel-6A measurements" ;
:ssalto_duacs_comment = "Sentinel-6A is the reference mission used for the altimeter inter-calibration processing" ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;
:summary = "SSALTO/DUACS Near-Real-Time Level-3 sea surface height measured by Sentinel-6A altimetry observations over Global Ocean." ;
:time_coverage_duration = "P23H36M42.630829S" ;
:time_coverage_end = "2024-04-30T23:59:19Z" ;
:time_coverage_resolution = "PT1.0S" ;
:time_coverage_start = "2024-04-30T00:22:36Z" ;
:title = "NRT Sentinel-6A (SAR mode) Global Ocean Along track SSALTO/DUACS Sea Surface Height L3 product with 1Hz (~7 km) posting rate" ;
}